

End-of-Unit Test — Thermo-S1 Part 1

Description of a system in thermodynamics · sections 1.A to 3.C

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 45 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 50 marks.**

Exercise 1. Unit conversions [1.A · 1.B · 2.A]

(3 marks)

Give each result exactly, without rounding.

(a) $V = 350 \text{ mL}$ in m^3 . (b) $\theta = -40^\circ\text{C}$ in kelvins. (c) $\chi_T = 5.0 \times 10^{-10} \text{ Pa}^{-1}$ in bar^{-1} .

(a) $V = 350 \times 10^{-3} \text{ L} = 0.350 \times 10^{-3} \text{ m}^3$	$V = 3.50 \times 10^{-4} \text{ m}^3$
(b) $T = \theta + 273.15 = -40 + 273.15$	$T = 233.15 \text{ K}$
(c) $1 \text{ Pa}^{-1} = 10^5 \text{ bar}^{-1}$	$\chi_T = 5.0 \times 10^{-5} \text{ bar}^{-1}$

Marking

①	0	1
②	0	1
③	0	1

Marking scheme

- ① 1 pt $3.50 \times 10^{-4} \text{ m}^3$.
- ② 1 pt 233.15 K (capital K, no degree sign).
- ③ 1 pt $5.0 \times 10^{-5} \text{ bar}^{-1}$.

Exercise 2. Extensive and intensive quantities — definitions [1.A]

(5 marks)

Define an extensive quantity and an intensive quantity, giving two examples of each. Show that the molar volume $V_m = V/n$ is intensive.

An extensive quantity is proportional to the size (amount of matter) of the system: if the system is doubled, it doubles, e.g. $V(2n) = 2V(n)$. Examples: mass m , volume V , amount of substance n , energy E .
An intensive quantity does not depend on the size of the system: it keeps the same value when the system is doubled. Examples: pressure p , temperature T , mass density ρ .
$V_m = V/n$: doubling the system doubles both V and n , so their ratio is unchanged. The quotient of two extensive quantities is intensive.

Marking

①	0	1	2
②	0	1	2
③	0	1	

Marking scheme

- ① **2 pts** Extensive: proportional to the size / doubles with the system (1 pt); two examples (1 pt).
 ② **2 pts** Intensive: independent of the size (1 pt); two examples (1 pt).
 ③ **1 pt** V_m intensive, justified.

Exercise 3. Systems and equilibrium [1.A · 1.B]

(3 marks)

A weather balloon (sealed, flexible envelope) is:

- ☐ open ☒ closed but not isolated
☐ isolated ☐ neither closed nor open

Why: no matter crosses the sealed envelope (closed), but heat and work are exchanged (the envelope is flexible and not insulating).

A gas in an insulating cylinder closed by a rigid lid is:

- ☒ isolated ☐ open
☐ closed but not isolated ☐ necessarily at 0 K

Why: sealed (no matter), rigid (no work) and insulating (no heat): Fig. 2 of the notes.

Thermodynamic equilibrium requires:

- ☐ only a steady state;
☐ only homogeneity;
☒ both: p and T uniform in space and constant in time, and equal to those of the surroundings;
☐ only thermal equilibrium.

Why: “the pressure and the temperature [must] depend neither on space (homogeneity) nor on time (steady state)”.

Exercise 4. State diagrams and coefficients [1.C · 2.B]

(4 marks)

The slope of an isotherm drawn as V versus p is:

- ☒ $\left(\frac{\partial V}{\partial p}\right)_T$ ☐ $\left(\frac{\partial V}{\partial T}\right)_p$ ☐ $\left(\frac{\partial p}{\partial T}\right)_V$ ☐ V/p

Why: on an isotherm T is fixed, and the slope of $V(p)$ is the partial derivative at fixed T .

For a gas, the sign of $\left(\frac{\partial V}{\partial T}\right)_p$ is:

- ☒ positive ☐ negative ☐ zero ☐ undetermined

Why: at fixed pressure a heated gas expands (Fig. 7).

The piezothermic coefficient is defined by:

- ☒ $\frac{1}{p} \left(\frac{\partial p}{\partial T}\right)_V$ ☐ $\frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_p$ ☐ $-\frac{1}{V} \left(\frac{\partial V}{\partial p}\right)_T$ ☐ $\frac{1}{T} \left(\frac{\partial T}{\partial p}\right)_V$

Why: Tab. 3: β describes p at fixed V . The others are α and χ_T .

The SI unit of χ_T is:

- ☒ Pa^{-1} ☐ K^{-1} ☐ Pa ☐ no unit

Why: $\chi_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p}\right)_T$ is the inverse of a pressure.

Exercise 5. A sealed container full of water [2.A · 2.C]

(6 marks)

A steel container, assumed rigid, is completely filled with liquid water and sealed. The coefficients of water are given below. (a) Write dV in terms of α , χ_T , dT and dp . (b) Deduce $\left(\frac{\partial p}{\partial T}\right)_V$ and compute it. (c) Estimate the pressure rise Δp when the water is heated by ΔT . (d) Compare with an ideal gas at $p = 1.0 \times 10^5$ Pa and $T = 300$ K, and conclude.

$\alpha = 2.0 \times 10^{-4} \text{ K}^{-1}$	$\chi_T = 5.0 \times 10^{-10} \text{ Pa}^{-1}$	$\Delta T = 5.0 \text{ K}$
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(a) $dV = \left(\frac{\partial V}{\partial T}\right)_p dT + \left(\frac{\partial V}{\partial p}\right)_T dp$ so $dV = \alpha V dT - \chi_T V dp$
(b) The container is rigid, so $dV = 0$: $\left(\frac{\partial p}{\partial T}\right)_V = \frac{\alpha}{\chi_T} = \frac{2.0 \times 10^{-4}}{5.0 \times 10^{-10}} = 4.0 \times 10^5 \text{ Pa K}^{-1}$ (4 bar per kelvin)
(c) $\Delta p \approx \frac{\alpha}{\chi_T} \Delta T = 4.0 \times 10^5 \times 5.0$ $\Delta p \approx 2.0 \times 10^6 \text{ Pa} = 20 \text{ bar}$
(d) Ideal gas at fixed V : $\left(\frac{\partial p}{\partial T}\right)_V = \frac{p}{T} = \frac{1.0 \times 10^5}{300} \approx 3.3 \times 10^2 \text{ Pa K}^{-1}$, about 10^3 times smaller. A liquid is almost incompressible, so even gentle heating in a closed full container builds up an enormous pressure: the container bursts. This is why heating circuits have expansion tanks.

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1
⑥	0	1

Marking scheme

- ① 1 pt $dV = \alpha V dT - \chi_T V dp$.
- ② 1 pt $dV = 0$, so $(\partial p / \partial T)_V = \alpha / \chi_T$.
- ③ 1 pt $4.0 \times 10^5 \text{ Pa K}^{-1}$.
- ④ 1 pt $\Delta p \approx 2.0 \times 10^6 \text{ Pa}$ (20 bar).
- ⑤ 1 pt Ideal gas: $p/T \approx 3.3 \times 10^2 \text{ Pa K}^{-1}$.
- ⑥ 1 pt Conclusion: ratio about 10^3 , danger of heating a full sealed container.

Exercise 6. Thermoelastic coefficients of the ideal gas [2.C · 3.A]

(4 marks)

 Starting from $pV = nRT$, compute α , χ_T and β for an ideal gas, and check that $\beta = \alpha/(p\chi_T)$.

$$V = \frac{nRT}{p}: \quad \alpha = \frac{1}{V} \frac{nR}{p}, \text{ so } \boxed{\alpha = \frac{1}{T}} \quad \chi_T = -\frac{1}{V} \left(-\frac{nRT}{p^2} \right), \text{ so } \boxed{\chi_T = \frac{1}{p}}$$

$$p = \frac{nRT}{V}: \quad \beta = \frac{1}{p} \frac{nR}{V}, \text{ so } \boxed{\beta = \frac{1}{T}}$$

$$\text{Check: } \frac{\alpha}{p\chi_T} = \frac{1/T}{p \cdot 1/p} = \frac{1}{T} = \beta. \checkmark$$

Marking

①
0 1
☐ ☐
②
0 1
☐ ☐
③
0 1
☐ ☐
④
0 1
☐ ☐

Marking scheme

- ① 1 pt $\alpha = 1/T$.
 ② 1 pt $\chi_T = 1/p$.
 ③ 1 pt $\beta = 1/T$.
 ④ 1 pt Relation checked.

Exercise 7. Historical laws [3.A]

(5 marks)

(a) An aerosol can (rigid) contains a gas at p_1 and θ_1 . It is left in the sun and reaches θ_2 . Which law applies? Compute the final pressure p_2 . (b) The bubble of a Cartesian diver, of volume V at pressure p , is compressed at constant temperature to the pressure p' . Compute its new volume V' .

(a) $p_1 = 3.0 \text{ bar}$	$\theta_1 = 27^\circ\text{C}$	$\theta_2 = 127^\circ\text{C}$
(b) $V = 1.0 \text{ cm}^3$	$p = 1.0 \text{ bar}$	$p' = 1.25 \text{ bar}$

(a) n and V are fixed: **Gay-Lussac's law**, $p/T = \text{const.}$, with $T_1 = 300 \text{ K}$ and $T_2 = 400 \text{ K}$:

$$p_2 = p_1 \frac{T_2}{T_1} = 3.0 \times \frac{400}{300} \text{ bar} \quad \boxed{p_2 = 4.0 \text{ bar}}$$

(b) n and T are fixed: **Boyle-Mariotte**, $pV = p'V'$:

$$V' = \frac{pV}{p'} = \frac{1.0 \times 1.0}{1.25} \text{ cm}^3 \quad \boxed{V' = 0.80 \text{ cm}^3}$$

Marking

①
0 1
☐ ☐
②
0 1
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③
0 1
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0 1
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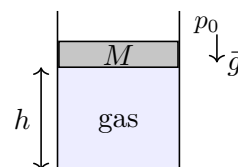
Marking scheme

- ① 1 pt Gay-Lussac identified (fixed n and V).
 ② 1 pt Temperatures in kelvins: 300 K and 400 K.
 ③ 1 pt $p_2 = 4.0 \text{ bar}$.
 ④ 1 pt $V' = pV/p'$ (Boyle-Mariotte).
 ⑤ 1 pt $V' = 0.80 \text{ cm}^3$.

Exercise 8. A piston under a weight [1.B · 3.B]

(7 marks)

A vertical cylinder of cross-section S is closed by a frictionless piston of mass M . The gas below it, treated as an ideal gas, fills a height h_1 at temperature T_1 . Outside, the pressure is p_0 .
 (a) Find the pressure p of the gas. (b) Find the amount of gas n . (c) The gas is heated slowly to T_2 , the piston remaining free. Find the new height h_2 and the displacement of the piston.



$$S = 100 \text{ cm}^2 \quad M = 20 \text{ kg} \quad g = 10 \text{ m s}^{-2} \quad p_0 = 1.0 \times 10^5 \text{ Pa}$$

$$h_1 = 25 \text{ cm} \quad T_1 = 300 \text{ K} \quad T_2 = 360 \text{ K} \quad RT_1 \approx 2.5 \times 10^3 \text{ J mol}^{-1}$$

- (a) Equilibrium of the piston: $pS = p_0S + Mg$, so $p = p_0 + \frac{Mg}{S}$, with
 $S = 1.0 \times 10^{-2} \text{ m}^2$:
 $\frac{Mg}{S} = \frac{20 \times 10}{1.0 \times 10^{-2}} = 2.0 \times 10^4 \text{ Pa}$ $p = 1.2 \times 10^5 \text{ Pa}$
- (b) $V_1 = Sh_1 = 1.0 \times 10^{-2} \times 0.25 = 2.5 \times 10^{-3} \text{ m}^3$
 $n = \frac{pV_1}{RT_1} = \frac{1.2 \times 10^5 \times 2.5 \times 10^{-3}}{2.5 \times 10^3} = 0.12 \text{ mol}$
- (c) The piston stays in equilibrium, so p does not change: the transformation is isobaric. Then $\frac{V_2}{T_2} = \frac{V_1}{T_1}$, and since $V = Sh$, $h_2 = h_1 \frac{T_2}{T_1} = 25 \text{ cm} \times \frac{360}{300}$
 $h_2 = 30 \text{ cm}$
 The piston rises by $\Delta h = 5.0 \text{ cm}$.

Marking

①		
0	1	2

②	
0	1

③	
0	1

④	
0	1

⑤			
0	1	2	

Marking scheme

- ① **2 pts** $p = p_0 + Mg/S$ (literal, 1 pt); $p = 1.2 \times 10^5 \text{ Pa}$ (1 pt).
 ② **1 pt** $V_1 = 2.5 \times 10^{-3} \text{ m}^3$.
 ③ **1 pt** $n = 0.12 \text{ mol}$.
 ④ **1 pt** Isobaric: V/T constant (the pressure is set by the piston).
 ⑤ **2 pts** $h_2 = 30 \text{ cm}$ (1 pt); $\Delta h = +5.0 \text{ cm}$ (1 pt).

Examiner's tip

$100 \text{ cm}^2 = 100 \times 10^{-4} \text{ m}^2 = 1.0 \times 10^{-2} \text{ m}^2$. With S in cm^2 the pressure would come out 10^4 times too small.

Exercise 9. Composition and density of air [1.A · 3.B]

(5 marks)

Dry air is modelled as a mixture of N_2 , O_2 and Ar with the mole fractions below. (a) Compute the molar mass M of air. (b) Show that the mass density of an ideal gas of molar mass M is $\rho = \frac{pM}{RT}$, and compute it for air at p and T given below.

$x_{\text{N}_2} = 0.78$	$x_{\text{O}_2} = 0.21$	$x_{\text{Ar}} = 0.01$
$M_{\text{N}_2} = 28 \text{ g mol}^{-1}$	$M_{\text{O}_2} = 32 \text{ g mol}^{-1}$	$M_{\text{Ar}} = 40 \text{ g mol}^{-1}$
$p = 1.0 \times 10^5 \text{ Pa}$	$T = 300 \text{ K}$	$RT \approx 2.5 \times 10^3 \text{ J mol}^{-1}$

(a) $M = \sum_i x_i M_i = 0.78 \times 28 + 0.21 \times 32 + 0.01 \times 40 = 21.84 + 6.72 + 0.40$
 $M \approx 29 \text{ g mol}^{-1}$

(b) $\rho = \frac{m}{V} = \frac{nM}{V}$, and $pV = nRT$ gives $\frac{n}{V} = \frac{p}{RT}$, so $\rho = \frac{pM}{RT}$.

With $M = 29 \times 10^{-3} \text{ kg mol}^{-1}$: $\rho = \frac{1.0 \times 10^5 \times 29 \times 10^{-3}}{2.5 \times 10^3} = \frac{2.9 \times 10^3}{2.5 \times 10^3}$
 $\rho \approx 1.2 \text{ kg m}^{-3}$

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1

Marking scheme

- ① 1 pt $M = \sum_i x_i M_i$.
- ② 1 pt $M \approx 29 \text{ g mol}^{-1}$.
- ③ 1 pt $\rho = pM/(RT)$, derived.
- ④ 1 pt M converted to kg mol^{-1} .
- ⑤ 1 pt $\rho \approx 1.2 \text{ kg m}^{-3}$.

Examiner's tip

In SI formulas the molar mass is in kg mol^{-1} . With $M = 29$ you would find $\rho = 1160 \text{ kg m}^{-3}$, denser than water: always sanity-check the result.

Exercise 10. A real gas at high pressure [3.C]

(8 marks)

One mole of nitrogen is compressed into a volume V at temperature T . (a) Compute the pressure p_{id} predicted by the ideal gas law. (b) Compute the Van der Waals pressure, evaluating separately its two terms. (c) Deduce the compressibility factor Z and interpret it using the Amagat diagram.

$$n = 1.0 \text{ mol} \quad V = 0.10 \text{ L} \quad T = 300 \text{ K} \quad RT \approx 2.5 \times 10^3 \text{ J mol}^{-1}$$

$$a = 0.14 \text{ Pa m}^6 \text{ mol}^{-2} \quad b = 4.0 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1} \quad (\text{rounded values for N}_2)$$

(a) $V = 0.10 \text{ L} = 1.0 \times 10^{-4} \text{ m}^3$ $p_{\text{id}} = \frac{nRT}{V} = \frac{2.5 \times 10^3}{1.0 \times 10^{-4}}$

$p_{\text{id}} = 2.5 \times 10^7 \text{ Pa}$ (250 bar)

(b) $p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$ $V - nb = 1.0 \times 10^{-4} - 0.40 \times 10^{-4} = 6.0 \times 10^{-5} \text{ m}^3$

repulsive term: $\frac{nRT}{V - nb} = \frac{2.5 \times 10^3}{6.0 \times 10^{-5}} \approx 4.2 \times 10^7 \text{ Pa}$ attractive term:

$a \frac{n^2}{V^2} = \frac{0.14}{1.0 \times 10^{-8}} = 1.4 \times 10^7 \text{ Pa}$

$p \approx 4.2 \times 10^7 - 1.4 \times 10^7$ $p \approx 2.8 \times 10^7 \text{ Pa}$ (280 bar)

(c) $Z = \frac{pV}{nRT} = \frac{p}{p_{\text{id}}} = \frac{2.8}{2.5}$ $Z \approx 1.1 > 1$

At this very high density the excluded volume (repulsion) wins over the attractions. This is the high-pressure part of the Amagat diagram, where $Z > 1$. (Compare progress test 3.C: CO₂ at 40 bar has $Z \approx 0.8$.)

Marking

①		
0	1	2
②		
0	1	
③		
0	1	
④		
0	1	2
⑤		
0	1	
⑥		
0	1	

Marking scheme

- ① **2 pts** $p_{\text{id}} = nRT/V$ with V in m³ (1 pt); $p_{\text{id}} = 2.5 \times 10^7 \text{ Pa}$ (1 pt).
- ② **1 pt** Van der Waals solved for p .
- ③ **1 pt** $V - nb = 6.0 \times 10^{-5} \text{ m}^3$.
- ④ **2 pts** Repulsive term $\approx 4.2 \times 10^7 \text{ Pa}$ (1 pt); attractive term $1.4 \times 10^7 \text{ Pa}$ (1 pt).
- ⑤ **1 pt** $p \approx 2.8 \times 10^7 \text{ Pa}$ and $Z \approx 1.1$.
- ⑥ **1 pt** Interpretation: $Z > 1$, the excluded volume dominates at high pressure.

Your score: / 50 → mark out of 20 = score $\times$ 20/50 = / 20

Diagnosis: where did you lose marks?

Ex.	Lecture section(s)	Redo
1	1.A, 1.B, 2.A: SI units, kelvins	Progress tests 1.A, 1.B, 2.A (Ex. 1)
2	1.A: extensive and intensive	Progress test 1.A
3	1.A, 1.B: systems, equilibrium	Progress tests 1.A (Ex. 2), 1.B
4	1.C, 2.B: diagrams, coefficients	Progress tests 1.C, 2.B
5	2.A, 2.C: α , χ_T , dV	Progress tests 2.A, 2.C
6	2.C, 3.A: coefficients of the ideal gas	Progress tests 2.C (Ex. 4), 3.A
7	3.A: historical laws	Progress test 3.A
8	1.B, 3.B: piston, $pV = nRT$	Progress tests 1.B (Ex. 4), 3.B (Ex. 4)
9	1.A, 3.B: composition, density	Progress tests 1.A (Ex. 4), 3.B
10	3.C: Van der Waals, Amagat	Progress test 3.C